

Task No.: Practice Tasks**Date: 08-01-2026**

Title: Develop an Android application for toast message, intent activity, services and Asynchronous tasks

Aim:

To develop an Android application for toast message, intent activity, services and asynchronous tasks.

Code:

MainActivity.kt

```
package com.example.practicetasks
```

```
import android.content.Intent
```

```
import android.os.AsyncTask
```

```
import android.os.Bundle
```

```
import android.widget.*
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    private lateinit var progressBar: ProgressBar
```

```
    private lateinit var txtAsyncStatus: TextView
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
        setContentView(R.layout.activity_main)
```

```
        val btnToast = findViewById<Button>(R.id.btnToast)
```

```
        val btnIntent = findViewById<Button>(R.id.btnIntent)
```

```
        val btnStartService = findViewById<Button>(R.id.btnStartService)
```

```
        val btnStopService = findViewById<Button>(R.id.btnStopService)
```

```
        val btnAsync = findViewById<Button>(R.id.btnAsync)
```

```
        progressBar = findViewById(R.id.progressBar)
```

```
        txtAsyncStatus = findViewById(R.id.txtAsyncStatus)
```

```

btnToast.setOnClickListener {
    Toast.makeText(this, "This is a Toast Message", Toast.LENGTH_SHORT).show()
}

btnIntent.setOnClickListener {
    startActivity(Intent(this, SecondActivity::class.java))
}

btnStartService.setOnClickListener {
    startService(Intent(this, MyService::class.java))
}

btnStopService.setOnClickListener {
    stopService(Intent(this, MyService::class.java))
}

btnAsync.setOnClickListener {
    MyAsyncTask().execute()
}
}

inner class MyAsyncTask : AsyncTask<Void, Int, String>() {

    override fun onPreExecute() {
        txtAsyncStatus.text = "AsyncTask Started"
        progressBar.progress = 0
    }

    override fun doInBackground(vararg params: Void?): String {
        for (i in 1..10) {
            Thread.sleep(500)
            publishProgress(i * 10)
        }
        return "AsyncTask Completed"
    }

    override fun onProgressUpdate(vararg values: Int?) {
        progressBar.progress = values[0]!!
        txtAsyncStatus.text = "Progress: ${values[0]}%"
    }
}

```

```

        override fun onPostExecute(result: String?) {
            txtAsyncStatus.text = result
        }
    }
}

```

SecondActivity.kt

```
package com.example.practicetasks
```

```

import android.os.Bundle
import android.widget.Button
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

```

```
class SecondActivity : AppCompatActivity() {
```

```

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_second)
    }

```

```
        Toast.makeText(this, "Opened using Intent", Toast.LENGTH_SHORT).show()
```

```
        val btnBack = findViewById<Button>(R.id.btnBack)
```

```

        btnBack.setOnClickListener {
            finish()
        }
    }
}

```

MyService.kt

```
package com.example.practicetasks
```

```
import android.app.Service
import android.content.Intent
import android.os.IBinder
import android.widget.Toast
```

```
class MyService : Service() {

    override fun onCreate() {
        super.onCreate()
        Toast.makeText(this, "Service Created", Toast.LENGTH_SHORT).show()
    }

    override fun onStartCommand(intent: Intent?, flags: Int, startId: Int): Int {
        Toast.makeText(this, "Service Started", Toast.LENGTH_SHORT).show()
        return START_STICKY
    }

    override fun onDestroy() {
        super.onDestroy()
        Toast.makeText(this, "Service Stopped", Toast.LENGTH_SHORT).show()
    }

    override fun onBind(intent: Intent?): IBinder? {
        return null
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">
```

```

<!-- TOP DETAILS -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center_horizontal"
    android:padding="16dp">

    <TextView
        android:text="Name: Yoga Shri P K"
        android:textStyle="bold"
        android:textSize="18sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TextView
        android:text="Roll No: 23BCS192"
        android:textStyle="bold"
        android:textSize="18sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>

<!-- CENTER TASKS -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:text="Practice Tasks"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <Button android:id="@+id/btnToast"

```

```
        android:text="Show Toast"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

<Button android:id="@+id/btnIntent"
        android:text="Open Second Activity"
        android:layout_marginTop="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

<Button android:id="@+id/btnStartService"
        android:text="Start Service"
        android:layout_marginTop="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

<Button android:id="@+id/btnStopService"
        android:text="Stop Service"
        android:layout_marginTop="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

<Button android:id="@+id/btnAsync"
        android:text="Run AsyncTask"
        android:layout_marginTop="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

<TextView
        android:id="@+id/txtAsyncStatus"
        android:text="AsyncTask Status"
        android:layout_marginTop="16dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

<ProgressBar
        android:id="@+id/progressBar"
        style="?android:attr/progressBarStyleHorizontal"
        android:max="100"
        android:progress="0"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"/>
    </LinearLayout>
```

```
</LinearLayout>
</ScrollView>
```

activity_second.xml

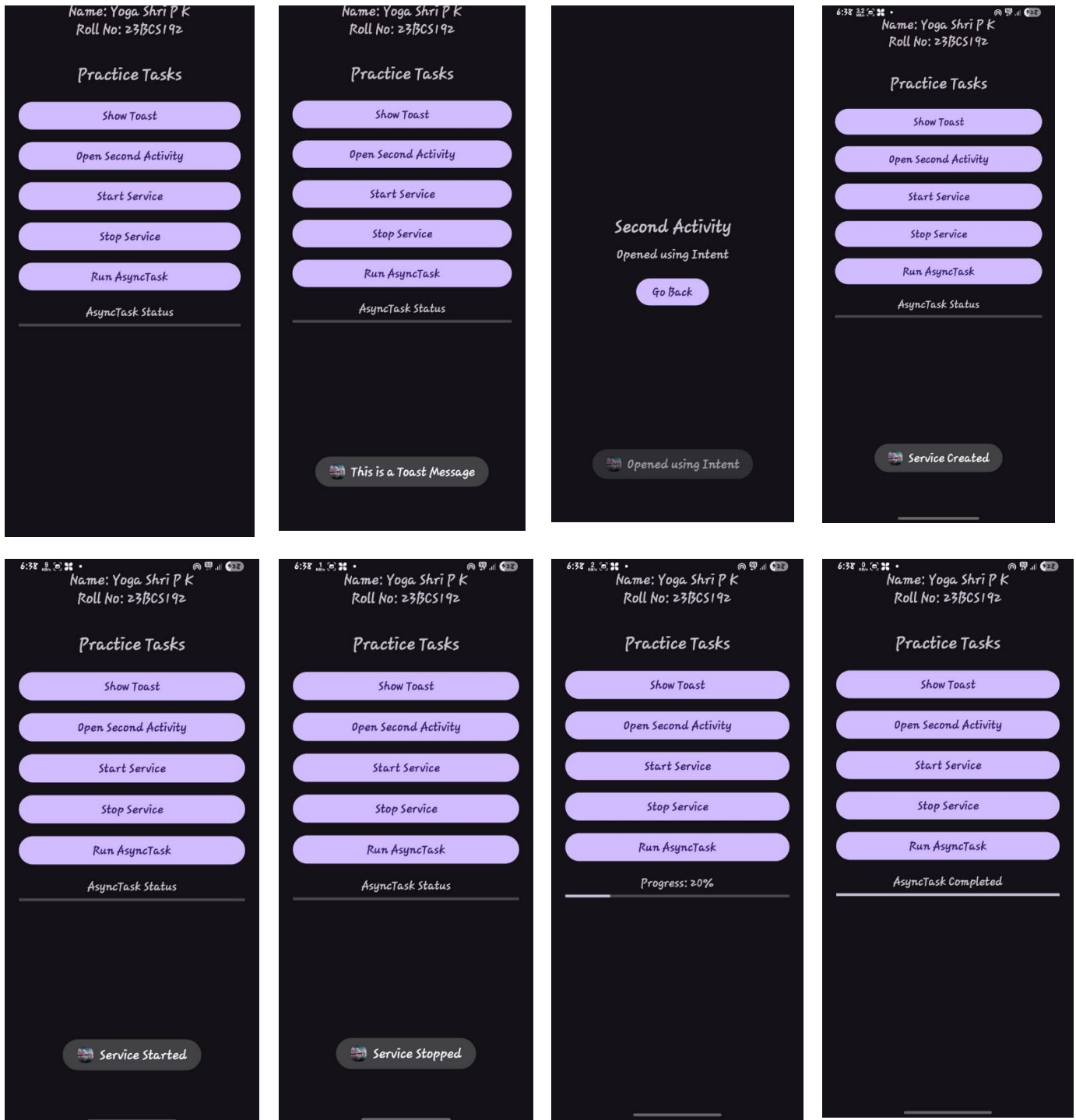
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Second Activity"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="12dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This screen is opened using Intent"
        android:textSize="16sp"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/btnBack"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go Back"/>
</LinearLayout>
```

Output:



Result:

An android application of containing toast message, indent, services and asynchronous tasks are executed successfully.